usb_joystick_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-06-04 | jlcpcb

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	12 (0 SMD, 0 THT, 12 other)
Nets	16
Traces	96 segments
Vias	22
Board Size	60.0 x 40.0 mm

Design Overview

Theory of Operation

USB Joystick Controller

USB game controller with analog joystick

Demonstrates autolayout functionality

Communication Interfaces

Protocol	Signals
USB	USB_D+, USB_D-

Power Architecture

Power Rails: +5V, GND, PWR_FLAG

Assembly Notes

1 fine-pitch component

- **Fine-pitch components**: 1 (U1)

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
100nF		4	C1, C2, C3, C4
Joystick		1	J2
USB-C		1	J1
Button		4	SW1, SW2, SW3, SW4
MCU		1	U1
16MHz		1	Y1

DRC Status

Metric	Count
Errors	9
Warnings	0
Blocking	9

Status: FAIL ### Violations by Type

Violation Type	Count
via_in_pad	6
clearance_segment_via	1
clearance_pad_via	1
connectivity	1
diffpair_clearance_intra	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 9 blocking DRC violations (via_in_pad (6), clearance_segment_via (1), clearance_pad_via (1))
- **[OPTIONAL]** Verify zone fill in KiCad: 1 nets appear incomplete but may be connected via zone fills
- **[OPTIONAL]** Verify zone fill in KiCad for 3 zone-connected nets

Routing Status

Metric	Value
Signal Net Completion	92.3% (12/13)
Overall Completion	93.8%
Complete Nets	15 / 16
Zone-Connected Nets	3
Incomplete Nets	1
Unconnected Pads	1

Zone-Connected Nets

- GND
- VBUS
- VCC

Unrouted Signal Nets

- USB_D+

Unrouted Signal Nets

- USB_D+

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.88 USD
Components (estimated)	~1.31 USD
Assembly (estimated)	~1.98 USD
Total (estimated)	**~4.17 USD**
Batch Quantity	5
Batch Total (estimated)	~20.87 USD